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Examples of nonunique factorizations

1. $\mathbb{Z}[\sqrt{-3}]$ ($d=-3$): $4 = 2 \cdot 2 = (1+\sqrt{-3})(1-\sqrt{-3})$.

2. $\mathbb{Z}[\sqrt{-5}]$ ($d=-5$): $6 = 2 \cdot 3 = (1+\sqrt{-5})(1-\sqrt{-5})$

3. $\mathbb{Z}[\sqrt{5}]$ ($d=5$): $4 = 2 \cdot 2 = (3+\sqrt{5})(3-\sqrt{5})$.

The factors here are all prime.

Q: Why shouldn't worry about associates?

A: e.g. $\frac{1+\sqrt{-3}}{2} = \frac{1}{2} + \frac{1}{2}\sqrt{-3} \notin \mathbb{Z}[\sqrt{-3}]$.

Division algorithm in $\mathbb{Z}[\sqrt{d}]$ for $d = -1, \pm 2, 3$.

Thm (DA in $\mathbb{Z}[\sqrt{d}]$) Given $\alpha, \beta \in \mathbb{Z}[\sqrt{d}]$ with $\beta \neq 0$,

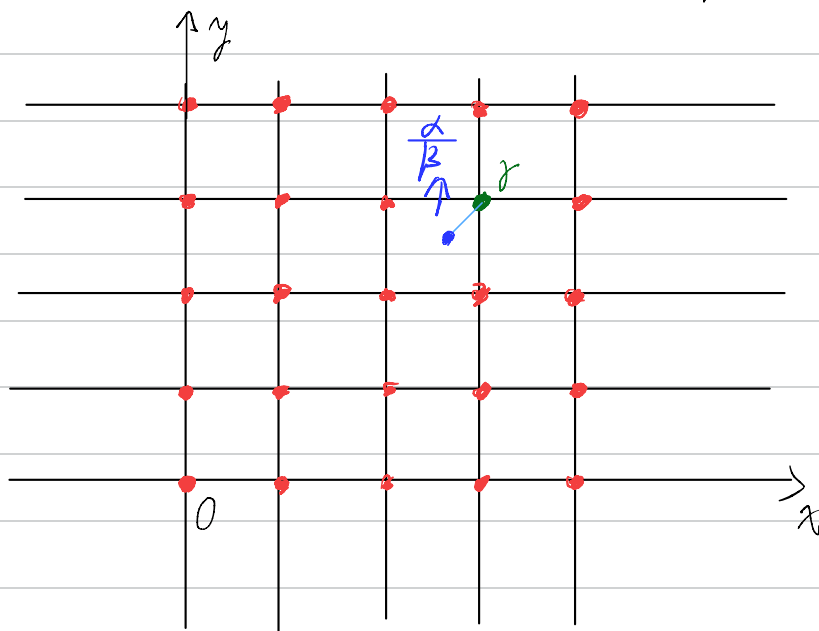
there are $r, s \in \mathbb{Z}[\sqrt{d}]$ s.t.

$$\alpha = \beta r + s, \quad |N(s)| < |N(\beta)|.$$

Idea: Take Gaussian integers $a+bi$ ($a, b \in \mathbb{Z}$)

corresponding to $d = -1$.

$a+bi$ $\longleftrightarrow$ lattice point (a, b)



$x_0 + iy_0$
||

Take $\frac{\alpha}{\beta} = x + iy \in \mathbb{C}$ and r to be an integer

closest to $\frac{\alpha}{\beta}$. Then $N(\frac{\alpha}{\beta} - r) = \underbrace{|x - x_0|^2}_{\leq \frac{1}{2}} + \underbrace{|y - y_0|^2}_{\leq \frac{1}{2}} \leq \frac{1}{2}$.

If $\delta = \alpha - \beta r$, then

$$N(\delta) = N(\beta \cdot (\frac{\alpha}{\beta} - r)) = N(\beta) \underbrace{N(\frac{\alpha}{\beta} - r)}_{\leq \frac{1}{2}} \leq \frac{1}{2} N(\beta).$$

Pf of DA: Suppose $d \in \{-1, \pm 2, 3\}$.

Consider $\frac{\alpha}{\beta} = x + y\sqrt{d}$. Choose $r \in \mathbb{Z}[\sqrt{d}]$ that is the

"closest" to $\frac{\alpha}{\beta}$ and write $r = x_0 + y_0\sqrt{d}$:

$$|x - x_0| \leq \frac{1}{2}, \quad |y - y_0| \leq \frac{1}{2}.$$

Then

$$|N(\frac{\alpha}{\beta} - r)| = |(\underbrace{x - x_0}_{\leq (\frac{1}{2})^2})^2 - d(\underbrace{y - y_0}_{\leq (\frac{1}{2})^2})^2|$$

$$\leq (x - x_0)^2 + |d|(y - y_0)^2 \leq \frac{1}{4} + \frac{1}{4}|d|.$$

So if $d = -1, \pm 2$, then

$$|N(\frac{\alpha}{\beta} - \gamma)| \leq \frac{1}{4} + \frac{1}{4}|d| \leq \frac{1}{4} + \frac{1}{4} \cdot 2 = \frac{3}{4} < 1.$$

When $d = 3$, then

$$N(\frac{\alpha}{\beta} - \gamma) = (x - x_0)^2 - 3(y - y_0)^2$$

$$\begin{aligned} \text{So } -3(y - y_0)^2 &\leq N(\frac{\alpha}{\beta} - \gamma) \leq (x - x_0)^2 \\ &\leq (\frac{1}{2})^2 = \frac{1}{4} \qquad \leq (\frac{1}{2})^2 = \frac{1}{4} < 1 \end{aligned}$$

In other words,

$$-1 < -\frac{3}{4} \leq N(\frac{\alpha}{\beta} - \gamma) \leq \frac{1}{4} < 1$$

$$\Rightarrow |N(\frac{\alpha}{\beta} - \gamma)| < 1.$$

Let $\delta = \alpha - \beta\gamma$. Then

$$|N(\delta)| = |N(\beta \cdot (\frac{\alpha}{\beta} - r))| = |N(\beta)| \underbrace{|N(\frac{\alpha}{\beta} - r)|}_{< 1} < |N(\beta)|. \quad \square$$

Remark. The proof actually shows

$$|N(\delta)| \leq \frac{3}{4} |N(\beta)|.$$

Next up, we prove, for $d = -1, \pm 2, 3$,

Prop ($\mathbb{Z}[\sqrt{d}]$ is a PID) Every ideal I of $\mathbb{Z}[\sqrt{d}]$ is of

the form $I = (\alpha) = \{\alpha\beta : \beta \in \mathbb{Z}[\sqrt{d}]\}$.

Pf. $I = \{0\}$ ✓

$I \neq \{0\}$: Take $\alpha \in I$ be nonzero element with

minimal $|N(\alpha)|$. If $\beta \in I$, then DA produces

$$\gamma, \delta \in \mathbb{Z}[\sqrt{d}] \quad \text{s.t.}$$

$$\beta = \alpha\gamma + \delta, \quad |N(\delta)| < |N(\alpha)|.$$

Since $\beta \in I$ and $\alpha \in I$,

$$\alpha\gamma \in I, \quad \delta = \beta - \alpha\gamma \in I.$$

By minimality of α and $|N(\delta)| < |N(\alpha)|$, we know

$$\delta = 0. \quad \text{So } \beta = \alpha\gamma, \text{ hence } \beta \in (\alpha). \text{ So } I \subseteq (\alpha).$$

$$\text{But } \alpha \in I \Rightarrow (\alpha) \subseteq I.$$

$$\text{Thus } I = (\alpha).$$

□

Defn (Greatest common divisor) Let $\alpha, \beta \in \mathbb{Z}[\sqrt{d}]$, not both

0. A gcd of α, β is some $\delta \in \mathbb{Z}[\sqrt{d}]$ s.t.

① $\delta \mid \alpha, \delta \mid \beta.$

② If $\varepsilon \mid \alpha, \varepsilon \mid \beta$, then $\varepsilon \mid \delta.$

Ex. If δ and ε are both gcds of α, β , then δ, ε are associates.

Notation. We will still write $\delta = \gcd(\alpha, \beta).$

Lemma (Bézout) Let $\alpha, \beta \in \mathbb{Z}[\sqrt{d}]$, not both 0. Let $\delta = (\alpha, \beta).$

Then the set

$$I_{\alpha, \beta} = \{\alpha r + \beta \varepsilon : r, \varepsilon \in \mathbb{Z}[\sqrt{d}]\}$$

is an ideal and $I = (\delta).$ In particular, there are

$\gamma, \varepsilon \in \mathbb{Z}[\sqrt{a}]$ s.t.

$$\delta = \alpha \gamma + \beta \varepsilon.$$